YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 1 OF 10 · 40 MINUTES

Think like a detective 像侦探一样思考

Detectives and scientists separate what they see (observations) from what they think it means (inferences), and trust evidence over opinion.

- I can sort statements into observations and inferences.
- I can write an observation and an inference about the scene, and say what evidence checks it.

forensic science 法庭科学 · detective 侦探 · evidence 证据 · crime scene 案发现场 · suspect 嫌疑人 · observation 观察 · inference 推断 · opinion 观点 · alibi 不在场证明

THE LESSON

Do Now: skills check	15	Seat fixed groups on arrival; read instructions once; 15 minutes alone, 5-minute warning
Do Now: interest survey	3	Collect skills checks; 6-question smiley survey; show Today slide (goals, key words)
Hook	4	Case alert: briefing and note on Slides 5 and 6; Song Ci story in one minute; open case files
Learn	4	Observation vs inference (glass); eye = observe, tap head = infer; O or I for 3 statements
Do	10	Say, show, check (Slide 9); scene map; Core (O or I, 8 statements), Stretch, Challenge
Think and Reflect	4	Meet the four suspects; thumbs vote (opinion); exit ticket: one O, one I; worksheet into file

Hands-on: No practical: paper scene map task, with an optional look-only scene table at the front.

Collect or assess: Skills Check baseline (out of 20) and survey into tracker; filed, not returned before Week 10. Exit ticket: O vs I.

SAFETY

- No practical hazards beyond normal classroom rules.
- Optional scene table: look only. No touching, tasting or smelling the evidence.
- Hot water for the optional scene liquid is teacher-only.

WATCH OUT

- Start the skills check as students sit; it takes the full 15 minutes. Never return papers before Week 10.
- Guesses get called observations ('The powder is sugar'). Ask: 'Can you see that, or are you guessing?'

Language: Frames: 'I observe ___ at square ___.' 'I infer ___ because ___.' Chinese pair talk welcome; Band 3 buddy for Band 1.

Next week: Week 2: fingerprints. Every case file needs its suspect cards. Keep pens A to D, strips and note for Week 3.

PREP FOR ONE CLASS

before the lesson about 220 min (155 could be done by someone else); after about 105 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Order the kit; plain soap and baking soda only if using the scene table 15 min · technician
- [ ] Book printing for Weeks 1 to 5; freeze Skills Check master until Week 10 15 min · teacher
- [ ] Print, fold and staple dictionaries x28 and logbooks x24 30 min · reprographics

THE WEEK BEFORE

- [ ] Print language pack (4 posters, cards) and worksheets x24 (48 sheets) 20 min · reprographics
- [ ] Laminate and cut role and language cards; cut frame cards; label class box 55 min · technician
- [ ] Pen pre-test (pens A to D, record, keep strips); write the note with pen B 25 min · teacher
- [ ] Print teacher copies; get allergy list (needed for Week 4 powders) 10 min · teacher
- [ ] Plan 6 groups of 4 (Band 1 beside a Band 3 buddy); make name cards 15 min · teacher
- [ ] Print scene table sign and 24 cm footprint at 100%; check with a ruler optional 5 min · technician

THE DAY BEFORE

- [ ] Collate 24 case files; order checks, cut surveys; posters up; set tables 35 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Make the scene liquid and set up the look-only scene table optional 15 min · teacher

AFTER THE LESSON

- [ ] Clear the scene table; keep the note and pens A to D for Week 3 optional 5 min · technician
- [ ] Mark the skills checks within the week (no half marks; accept Chinese) 70 min · teacher
- [ ] Enter tracker, file papers; read exit tickets; check case files in box 35 min · teacher

PRINT

- Skills Check x24 (3 sheets each, double-sided; length to confirm)
- Worksheet x24 (2 sheets each); interest survey x24 (half A4, cut)
- Briefing (double-sided), scene map, suspect cards x24 (1 sheet each)
- Case file cover x24, evidence log x24 (A4 card if possible)
- Dictionary x28 (A5 booklet, 7 sheets each, 90 gsm); logbook x24 (A5)
- Wall: A3 scene map, 3 A3 language posters, A4 frame poster
- Frame cards x6 sets (4 sheets each); role cards x6 sets (1 sheet each)
- Classroom language cards x30 (A5, double-sided, laminated)
- Teacher copies: kit card, mark scheme, pre-test record
- Scene table sign (A5), true-size footprint (A4 at 100%) (optional)

GET OR MAKE (NEW THIS LESSON)

- Black washable felt-tips x8 (4 brands, 2 of each)
- Coffee filters (4 strips, 2 x 10 cm), cups x4, water
- Masking tape x1 roll
- White A4 card x49 (covers, evidence logs, note)
- A4 folders x24 (collated into case files)
- Pencils x24
- Laminating pouches: A4 x6, A5 x30
- Class box x1 (a section per group)
- Class assessment folder x1 (may already exist)
- Name cards (number not stated; 24 assumed)
- Make: pen pre-test strips x4, display note
- Scissors or guillotine, laminator, ruler and pencil
- Scene table: glass, dark tray, saucer (optional)
- Bar soap, baking soda, grater, kettle, jug (optional)

Have ready from earlier weeks: Nothing.

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Fingerprints 指纹档案

Scientists classify things and count how common each group is; how common a feature is decides how strong it is as evidence.

- I can take my own fingerprint and classify it as a loop, whorl or arch.
- I can use class data in a bar chart to explain why a common fingerprint type is weak evidence.

fingerprint 指纹 · **pattern** 纹型 · **loop** 箕形纹 · **whorl** 斗形纹 · **arch** 弓形纹 · **classify** 分类 · **tally** 划记计数 · **bar chart** 条形图 · **common / rare** 常见的／罕见的

THE LESSON

Do Now	4	Four questions from the suspect cards; check roles; log row 1 if not done; Today slide
Hook	3	Fingerprint facts and an ancient Chinese finger-mark contract; fingertips under magnifiers
Learn	7	Loop, whorl, arch with gestures; how common each is; teacher demos the pencil-and-tape method
Do	18	Take prints (5 min), classify (3), tally and bar chart (6), Fingerprint File at minute 14 (4)
Think and Reflect	8	How strong is the fingerprint evidence? Pair talk; evidence log row 2, tracker; exit ticket

Hands-on: Students take own prints with soft pencil and tape onto a record card, classify with a magnifier, add a tally.

Collect or assess: Record card; tally and bar chart; evidence log row 2 and suspect tracker; exit ticket on common vs rare evidence.

SAFETY

- Graphite on fingers: wash hands after taking fingerprints; wipe tables.
- No pencils or tape in mouths.
- Take only your own fingerprints.

WATCH OUT

- A matching type is not proof: the Think answer is 'sometimes'; 3 of 4 suspects have loops.
- Smudges: lots of graphite, gentle press, no sliding. One tally per student, right index finger only.

Language: Frames on tables; Chinese talk and answers welcome. Band 1 beside a Band 3 buddy. Traps: 'loop', 'arch', 'tally'.

Next week: Week 3 (chromatography): keep tape rolls; this week cut filter strips 2 cm by 10 cm, make pen B ink 'N' strips

PREP FOR ONE CLASS

before the lesson about 50 min (35 could be done by someone else); after about 21 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Order pencils 2B to 4B x24, clear tape x6 rolls, A4 white card x12 sheets 10 min · technician
- [ ] Book printing: worksheets x24 (48 sheets), kit card x1 (A5) 5 min · reprographics

THE WEEK BEFORE

- [ ] Gather prep room kit; check visualiser works and sink has soap 10 min · technician
- [ ] Print record cards on white card (12 sheets, cut) and Fingerprint File x7 10 min · technician
- [ ] Check each case file has its suspect cards and evidence log 5 min · teacher
- [ ] Practise one pencil-and-tape print before the demo (optional) optional 5 min · teacher

THE DAY BEFORE

- [ ] Plan seating: each Band 1 student beside a Band 3 language buddy 5 min · teacher
- [ ] Tape scene print onto a drinking glass for display (optional) optional 2 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Draw tally table as Slide 9; set out 6 trays, table kit; visualiser on 5 min · teacher

AFTER THE LESSON

- [ ] Clean-up: cards into case files, kit in trays, wipe tables, wash hands 3 min · students
- [ ] Read exit tickets, check some record cards; files, logbooks to class box 10 min · teacher
- [ ] Between classes: count pencils and tape; refill trays; replace marked Files 5 min · technician
- [ ] After last class: return magnifiers, trays, cloths; keep 6 tape rolls 3 min · technician

PRINT

- Worksheet x24 (2 sheets each, pages 1 and 2 double-sided)
- Fingerprint File x7 (A4 single-sided: 6 groups + 1 display)
- Record cards x24 on A4 white card (2 per sheet, 12 sheets, cut)
- Teacher copy: kit card (A5)
- Extra Fingerprint File x1 to cut for the display glass (optional)

GET OR MAKE (NEW THIS LESSON)

- Soft pencils 2B to 4B x24 (4 per group)
- Clear sticky tape x6 rolls (1 per group)
- A4 white card x12 sheets (record cards)
- Scrap paper x24 sheets (one side used is fine)
- Magnifying glasses x12 (2 per group)
- Trays x6 (1 per group)
- Cloths for wiping tables (number not stated)
- Paper towels or wet wipes (a few per group)
- Visualiser x1 (may already be in the room)
- Class tally table drawn on the board
- Drinking glass with scene print taped on (optional)

Have ready from earlier weeks: Class box, Case files x24 (with suspect cards), Scientist's Logbooks x24, Writing pencils x24, Role cards incl. Language buddy x6 sets, Sentence frame card sets x6, My Science Dictionary x24, Classroom language cards x24

Chinese: draft, needs native-speaker check

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Who wrote the note? 纸条是谁写的?

Black ink is a mixture of dyes. Chromatography separates them into a pattern we match against reference samples, our controls.

- I can run a paper chromatogram and use reference samples to match the note ink to a pen.
- I can explain what the match shows and what it does not show.

chromatography 色谱法 · **chromatogram** 色谱图 · **mixture** 混合物 · **dye** 染料 · **solvent** 溶剂 · **dissolve** 溶解 · **reference sample** 参照样本 · **baseline** 基线 · **solvent front** 溶剂前沿 · **Rf value (Stretch)** 比移值（Rf值）

THE LESSON

Do Now	4	Fingerprint flashback: patterns 1 to 3, who was ruled out, strong or weak evidence; Today slide
Hook	3	Show the note: 'Is black ink just black?' Predict, then see a pre-test strip split into colours
Learn	7	How chromatography works; reference samples as controls; model one strip; method cloze; rules
Do	18	Set up 5 strips (5 min); Core a as they run (7 min); strips out at 12 min; match N; Stretch: Rf
Think and Reflect	8	Does the match prove who wrote it? Pair talk, frame; evidence log row 3, tracker; exit ticket

Hands-on: Groups of 4 run 5 chromatograms (pens A to D and note strip N) in 1 cm of water for about 7 min, then match N.

Collect or assess: Results and match tables (Stretch: Rf); evidence log row 3 and suspect tracker 'Ink'; exit ticket on reference samples.

SAFETY

- Water in the cups: do not drink it. Wipe up spills straight away.
- Felt-tip ink stains clothes: keep caps on when not in use.
- Wash hands after the practical.

WATCH OUT

- Water above the baseline is the most common failure: check every cup (1 cm water, baseline 2 cm up).
- 'The pen matches, so that person wrote it': two suspects carry pen B and pens can be borrowed.

Language: Frame: 'This sample matches ___ because ___.' Chinese welcome in pair talk. Traps: solvent/solution, dye/die.

Next week: Keep worksheets for Week 4 box d, spares for Weeks 7 to 8. This week, for Week 4: allergy check, powder pots.

PREP FOR ONE CLASS

before the lesson about 82 min (60 could be done by someone else); after about 23 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Buy 24 group pens (same 4 brands as pre-test); check filters for 44 strips 10 min · technician
- [ ] Print the Teacher Pre-test Record (1 copy) 2 min · teacher
- [ ] Pen pre-test: choose pen B, label A to D, fill in record, write the note 20 min · teacher

THE WEEK BEFORE

- [ ] Book printing: 24 worksheets (48 sheets) and the kit card 5 min · reprographics
- [ ] Label 24 group pens A to D with masking tape; make up 6 sets of 4 10 min · technician
- [ ] Print and laminate 6 lab cards in A5 pouches 10 min · technician
- [ ] Cut 44 filter strips (2 cm by 10 cm); make 8 N strips with pen B 10 min · technician

THE DAY BEFORE

- [ ] Optional, only if one pen set: pre-dot 24 strips A to D (optional) 10 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out 6 kit trays, table cards, and teacher desk items 10 min · technician

SET-UP BY STUDENTS IN THE LESSON

- [ ] Groups set up by role: baselines, dots, water to 1 cm, label cups 5 min · students

AFTER THE LESSON

- [ ] Clean-up: empty cups, dry strips flat, count pens, wipe tables, wash hands 3 min · students
- [ ] Check 24 pens, rinse cups and jugs, store strips for Week 4, reset trays 10 min · technician
- [ ] Read exit tickets; note who needs the Week 4 Do Now recap 10 min · teacher

PRINT

- Worksheet x24 (2 sheets each, double-sided)
- Chromatography lab card x6 (A5, single-sided, laminated)
- Teacher copies: kit card (A5), Pre-test Record (A4, print in Week 1)

GET OR MAKE (NEW THIS LESSON)

- Black washable felt-tips x24 (6 sets, brands A to D)
- Small clear plastic cups x30
- Kit pencils x30 (5 per group)
- Rulers, 30 cm x6
- Small jugs of water x6
- Trays x6
- Coffee-filter strips x44 (2 cm by 10 cm, cut fresh)
- Note ink strips N x8 (made with pen B)
- Paper towels x12 (2 per group)
- Laminating pouches, A5 x6
- Cloths for wiping tables (requisition: 6)
- Guillotine or scissors for cutting strips
- Paper clips, if not using tape (optional)
- Pre-dotted strips A to D x24, if one pen set (optional)

Have ready from earlier weeks: Class box with case files and logbooks, Role card and frame card sets x6, Dictionaries and writing pencils x24, Teacher pens A to D x8 (pen B), Dried pre-test strips A to D (key), Display note written with pen B, Sticky tape rolls x6 (Week 2), Masking tape roll (Week 1)

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The white powder 白色粉末之谜

Testing every sample the same way and recording results in a grid lets us identify an unknown against known samples; a grid can become a key.

- I can test five known powders and the scene powder systematically and record results in a grid.
- I can use my results to identify the scene powder and turn the grid into a flowchart key.

systematic 系统的 · **soluble** 可溶的 · **insoluble** 不溶的 · **fizz** 冒气泡 · **react** 反应 · **starch** 淀粉 · **iodine solution** 碘液 · **crystal** 晶体 · **results grid** 结果记录表 · **identification key** 检索表

THE LESSON

Do Now	4	Recap questions; stick dry Week 3 chromatograms into Week 3 worksheet; goals. Slides 2 to 3
Hook	3	Hold up Evidence 3 bag: which of five canteen powders? Hands-up vote, numbers on board. Slide 4
Learn	7	Three tests and positive results; model one powder; be systematic; goggles on. Slides 5 to 7
Do	18	Test 5 knowns, scene powder last; Recorder fills grid (12 min); key (6 min). Slides 8 to 10
Think and Reflect	8	Name powder; linked suspects; evidence log row 4; confidence vote; exit ticket. Slides 11 to 12

Hands-on: Groups of 4 test five known white powders and the scene powder: magnifier, water, vinegar, iodine (optional).

Collect or assess: Results grid and flowchart key; evidence log row 4, suspect tracker and confidence row; exit ticket; logbook if time.

SAFETY

- Goggles on before vinegar or iodine; iodine stains: teacher or Safety Officer only
- Wheat or corn allergy: no flour or cornflour handling; give them the Recorder role
- Never taste powders or liquids; do not blow or shake powders; wash hands after

WATCH OUT

- Weak fizzing gets missed: watch each well for 10 seconds; one clean spoon per powder.
- Many groups will name Dr Huang: stay neutral, do not confirm or deny; ask how sure they are.

Language: Frame: 'The scene powder matches ___ because it ___.' Soluble vs solvent trap; fizz, starch new. Band 1: buddy, Measurer

Next week: Week 5: print checkpoint papers; laminate colour pH chart; red cabbage indicator day before (25 min, chill).

PREP FOR ONE CLASS

before the lesson about 70 min (35 could be done by someone else); after about 34 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book printing: 24 worksheets (48 sheets), 6 lab cards, kit card, A3 poster 5 min · reprographics
- [ ] Get or buy powders and vinegar; order pots, bags, sticks, card, labels 10 min · teacher

THE WEEK BEFORE

- [ ] Ask prep room for goggles, tiles, cups, droppers, trays, iodine if used 5 min · technician
- [ ] Allergy check; fill and label 6 powder sets, 6 Evidence 3 bags; fizz test 15 min · teacher
- [ ] Laminate and trim 6 powder testing lab cards (A5 pouches) 10 min · technician

THE DAY BEFORE

- [ ] Find Week 3 chromatograms and worksheets; plan roles and seating 10 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Pour vinegar, fill 36 water cups, set out 6 trays and table materials 15 min · technician

AFTER THE LESSON

- [ ] Clean up: empty cups, rinse tiles, wipe tables, wash hands, pack class box 4 min · students
- [ ] Wash kit; store iodine; refill powder pots, bags and vinegar for next class 15 min · technician
- [ ] Read exit tickets and grids; note Week 5 recap and unfinished Stretch keys 10 min · teacher
- [ ] Put up A3 key poster once every class has done Week 4 5 min · teacher
- [ ] Optional: keep unopened Evidence 3 bags, pots, black card for Weeks 7 and 8 optional 2 min · technician

PRINT

- Worksheet x24 (3 pages on 2 A4 sheets; pages 1 and 2 back to back)
- Powder testing lab card x6 (A5, laminated)
- Identification key poster x1 (A3, wall from Week 5)
- Teacher copy: kit card x1 (A5)

GET OR MAKE (NEW THIS LESSON)

- Salt, sugar, cornflour, plain flour: about 12 tsp each
- Baking soda about 30 tsp; white vinegar about 180 ml
- Small lidded pots x30; small zip bags x6
- Stirring sticks x36; spoons or lolly sticks x36
- Black card x6; sticky labels x42
- Laminating pouches, A5 x6
- Safety goggles x24
- Spotting tiles (12 wells) x6
- Small clear plastic cups x42
- Droppers (plastic pipettes) x6
- Iodine solution dropper bottles x6 (optional)
- Teacher's measuring teaspoon
- Paper towels and tap water (lab supply)
- Make: known powder sets x6 (5 labelled pots each)
- Make: Evidence 3 bags x6 (about 3 tsp baking soda)
- Teacher demo set (optional)

Have ready from earlier weeks: Class box: case files, logbooks, Week 3 worksheets and dry chromatograms, Role cards and frame card sets x6, Science Dictionaries and pencils x24, Trays and magnifiers x6; cloths, Clear sticky tape (Weeks 2 and 3)

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The mystery liquid 神秘液体

An indicator's colour shows if a liquid is acid, neutral or alkali; reference samples place it on the pH scale, but not which substance it is.

- I can use indicator and reference samples to classify the scene liquid and estimate its pH.
- I can notice evidence that does not fit and suggest explanations that could be checked.

indicator 指示剂 · **acid** 酸性 · **alkali** 碱性 · **neutral** 中性 · **pH scale** pH值（酸碱度） · **colour chart** 比色卡 · **reference sample** 参照样本 · **consistent** 一致的

THE LESSON

Do Now (Checkpoint)	10	Papers on desks; exactly 10 minutes on the timer; collect, then show the Today slide
Hook	2	Demo: vinegar turns indicator pink, baking soda solution blue-green; students call colours
Learn	5	Acid, neutral, alkali, pH scale and colour chart; safety slide; goggles on; copy colour key
Do	15	Groups add 10 drops of each liquid to indicator cups; record colour, type, pH; match suspects
Think and Reflect	8	Why does the liquid not fit? Pair talk; sticky note idea; evidence log row 5; exit ticket

Hands-on: Groups test 5 reference liquids and the scene liquid in red cabbage indicator; read pH; match to suspects.

Collect or assess: Checkpoint (12 marks) into tracker column Q; results table; evidence log row 5 and tracker; exit ticket; sticky notes.

SAFETY

- Eye splashes: goggles on before opening any liquid; rinse eyes with plenty of water.
- Never taste or drink any liquid. Hot water and the kettle: teacher only.
- Wipe spills straight away; check allergy list (lemon juice); wash hands after.

WATCH OUT

- Test the scene liquid first: it must turn blue-green, not purple. Bottled water for indicator and water.
- Keep checkpoint to exactly 10 minutes. Do not explain why the liquid does not fit or confirm a suspect.

Language: Chinese welcome in talk and answers. Band 1 beside language buddy. Traps: spelling 'alkali'; 'neutral' not 'natural'.

Next week: Week 6 (footprints, incoming teacher): keep Parking Board up; print scene footprint at 100% (24 cm).

PREP FOR ONE CLASS

before the lesson about 84 min (30 could be done by someone else); after about 75 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Buy red cabbage (1/4 head), lemon juice, bar soap, bottled water; check kit 10 min · teacher

THE WEEK BEFORE

- [ ] Check the allergy list for lemon juice; plan for any student affected 2 min · teacher
- [ ] Print: checkpoint x25, worksheet x24, teacher copies, colour pH charts x7 10 min · reprographics
- [ ] Laminate and trim the 6 A5 pH colour charts 10 min · technician

THE DAY BEFORE

- [ ] Label 72 cups: 36 indicator, 30 reference, 6 Evidence 4 (optional) optional 15 min · technician
- [ ] Make red cabbage indicator (hot water, teacher only); strain, label, fridge 25 min · teacher
- [ ] Put up Parking Board header, A3 pH chart and Week 4 powder key poster 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Make soap, baking soda, scene liquids; fill, label, test cups; papers out 15 min · teacher
- [ ] Set out 6 trays, Hook demo, sticky notes, case files and cards 5 min · technician
- [ ] Set incoming teacher's seat: lesson plan and a checkpoint paper 2 min · teacher

AFTER THE LESSON

- [ ] Clean-up: liquids down sink, goggles away, wipe tables, wash hands 3 min · students
- [ ] Rinse cups and droppers or recycle; leftover indicator in fridge (3 days) 10 min · technician
- [ ] Photograph Parking Board; read exit tickets; book Week 6 walkthrough 17 min · teacher
- [ ] Mark checkpoint out of 12, enter tracker column Q, then write handover note 45 min · teacher

PRINT

- Worksheet x24 (2 sheets each, pages 1 and 2 double-sided)
- Checkpoint paper x25 (1 sheet, double-sided; 1 for incoming teacher)
- pH colour chart x6 (A5 landscape, colour, laminated)
- pH colour chart x1 (A3 landscape, colour if possible, for the wall)
- Ideas Parking Board header x1 (A4 landscape)
- Lesson plan (one page) x1 for the incoming teacher's seat
- Teacher copies: kit card (A5, 2 sides), mark scheme, handover form

GET OR MAKE (NEW THIS LESSON)

- Small clear plastic cups x74 (72 on trays, 2 for demo)
- Droppers x36 (6 per group)
- Safety goggles x24, trays x6
- Sticky labels x73, sticky notes x24
- Plain white A4 paper x6, A5 laminating pouches x6
- Kettle, heatproof jug, sieve, clean bottle (1 each)
- Kitchen tools: knife, board, grater, spoons, jugs
- Red cabbage x1/4 head
- Lemon juice 60 ml; vinegar 60 ml plus a little for demo
- Plain bar soap x1 (not washing-up liquid), baking soda
- Bottled or purified water 560 ml
- Cloths and paper towels (number not stated)
- Timer, camera or phone, fridge space, board wall space
- Make: indicator 500 ml, soap solution 200 ml
- Make: baking soda solution; scene liquid about 60 ml
- Rinse water cups x6, if 2 droppers per group (optional)

Have ready from earlier weeks: Class box, Case files x24, Scientist's Logbooks x24, My Science Dictionary x28 (4 spare), Role cards incl. language buddy x6 sets, Sentence frame card sets x6, poster, Week 4 powder key poster (A3), Pencils x24

Chinese: draft, needs native-speaker check

YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 6 OF 10 · 40 MINUTES

Footprints and height 脚印与身高

Linked measurements (a correlation) let a scatter graph and line of best fit estimate one from the other; people vary, so give a range.

- I can measure foot length and height, plot a scatter graph and draw a line of best fit.
- I can estimate the footprint owner's height as a range from my graph and name suspects who fit.

foot length 脚长 · **height** 身高 · **data** 数据 · **scatter graph** 散点图 · **correlation** 相关性 · **line of best fit** 最佳拟合线 · **estimate** 估计 · **range** 范围

THE LESSON

Do Now	4	New case leader intro; 3 Week 5 recap questions from the evidence log; Parking Board; goals
Hook	3	True-size footprint by a metre stick: can it tell height? Vote short, medium or tall
Learn	7	Scatter graph, correlation, line of best fit; read off a range; model measuring safely, kindly
Do	18	Measure in pairs, class data table, plot scatter graph, line of best fit, read off at 24 cm
Think and Reflect	8	Which suspects fit the range? Pair talk; evidence log row 6, tracker; exit ticket with vote

Hands-on: Pairs measure foot length (shoes off, 0.5 cm) and height (1 cm), plot a scatter graph and read off at 24 cm.

Collect or assess: Scatter graph and estimate (rubric strand 4); evidence log row 6, tracker, W6 confidence; exit ticket with Week 7 vote

SAFETY

- Shoes off, socks on: shoes under desks, back on before anyone leaves their seat
- No standing on chairs or tables; never swing metre sticks; retract tapes slowly
- Measuring is voluntary: numbers not names, no comments on size; sample data allowed

WATCH OUT

- Measuring slow or under 16 measured: switch all to the sample data card; never mix data on one graph.
- Joining dots, starting at zero or one number: ask 'As many crosses above as below?' Insist on a range.

Language: Frame: 'The person was probably between ___ cm and ___ cm tall.' Say 'height', not 'high'. Chinese answers count.

Next week: Exit tickets: give each group one option (Soil, Paper, Ink 2, Powder 2), sets Week 8 kit; collect Week 7 soil

PREP FOR ONE CLASS

before the lesson about 103 min (43 could be done by someone else); after about 28 min (some times estimated)

THE WEEK BEFORE

- [ ] Meet current teacher: handover, tracker, freezer decision, Weeks 6, 7 kit 30 min · teacher
- [ ] Read Case Bible Sections 4 to 6 (suspects' heights and foot lengths) 20 min · teacher
- [ ] Send print job (rebuilt designs): worksheets, data cards, signs, kit card 5 min · reprographics
- [ ] Print footprint at 100% (not fit to page): 6 per class plus 1; check 24 cm 5 min · technician
- [ ] Laminate the 7 sample data cards in A5 pouches (optional) optional 10 min · technician
- [ ] Borrow 6 hardbacks; collect rulers, metre sticks, trays; sharpen 24 pencils 20 min · technician
- [ ] Find the Week 5 Ideas Parking Board or its photo for the Do Now 5 min · teacher

THE DAY BEFORE

- [ ] Set up 2 or 3 height stations (chart or metre stick, sign); check the scale 8 min · technician
- [ ] Check the slide 9 data table can be typed into while projected 3 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Decide: class's own data or sample data (sample if class small) 2 min · teacher
- [ ] Set out group trays, case files, role and frame cards; Hook kit at front 5 min · technician

AFTER THE LESSON

- [ ] Clean-up: shoes on, kit counted, data cards collected, footprints filed 3 min · students
- [ ] Read exit tickets: record Week 7 option choices, give each group one option 15 min · teacher
- [ ] Last class: return kit and books, store data cards; leave charts if needed 10 min · technician

PRINT

- Worksheet x24 (2 sheets each, double-sided)
- Scene footprint x6 per class, A4 at 100%, check 24 cm (1 per group)
- Sample class data card x6, A5, laminated if possible (1 per group)
- Height station sign x3, A4 (1 per station)
- Teacher copies: kit card (A5), sample data card, Hook footprint

GET OR MAKE (NEW THIS LESSON)

- Rulers, 30 cm x12 (2 per group)
- Metre sticks or tape measures x7 (6 groups, 1 Hook)
- Hardback books x6 (borrow from library or staff room)
- Paper height charts, 130 to 190 cm x3 (buy ready-made)
- Masking tape x1 roll
- Up to 3 metre sticks for upright stations (optional)
- Laminating pouches, A5 x7 (optional)
- Make: height stations x3 (chart, masking tape, sign)

Have ready from earlier weeks: Class box, Case files x24, Scientist's Logbooks x24, My Science Dictionaries x24, Role card and frame card sets x6 each, Group trays x6, Sharp pencils x24 (4 per group), Ideas Parking Board (Week 5) or photo

YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 7 OF 10 · 40 MINUTES

Design your own test 设计你的检测

A good forensic test has a clear question, tests the scene sample exactly like the references, changes one thing and is planned to be safe.

- I can write a testable question and name what we change, observe or measure, and keep the same.
- I can plan a safe method with reference samples, repeats, a risk table and a results table.

testable question 可检验的问题 · **independent variable** 自变量 · **dependent variable** 因变量 · **control variable** 控制变量 · **fair test** 公平测试 · **reference sample** 参照样本 · **method** 实验步骤 · **prediction** 预测 · **hazard** 危险源 · **risk** 风险

THE LESSON

Do Now	4	Three recap questions on whiteboards or in the logbook; Today slide with goals and key words
Hook	3	Show sealed Evidence 6 and 7 bags; hand out option cards; Reporters read the question aloud
Learn	7	Six planning steps modelled with the Week 4 powder test; one worked risk-table row (vinegar)
Do	18	Groups plan on the plan sheet; checkpoints at minutes 6 and 12; results table and kit request
Think and Reflect	8	Pick the fairer soil method; plan swap (skip if short); exit ticket; collect sheets at the door

Hands-on: No practical: groups plan their Week 8 test (Soil, Paper, Ink 2 or Powder 2) and only look at sealed bags.

Collect or assess: Plan sheet (rubric strands 1 and 2) signed off at checkpoints 1 and 2; exit ticket; kit requests; evidence log row 7.

SAFETY

- No practical work: evidence bags stay sealed and go back to the teacher.
- Do not open, smell or taste any sample.
- Teacher checks every group's risk table before any Week 8 practical.

WATCH OUT

- Groups change two things, list kit instead of a method, or skip repeats: insist on the 'We change' frame.
- Evidence 6 must come from flowerbed A; Evidence 7 from the lab's own roll, arrow marked before crumpling.

Language: Frames: 'We change / observe / keep ___'. Hazard is the thing, risk the chance. Band 1: language buddy, Core b) first.

Next week: Week 8 lab day: commented plan sheets back in the Do Now; keep option cards and bags; trays from kit requests.

PREP FOR ONE CLASS

before the lesson about 50 min (10 could be done by someone else); after about 43 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book printing: plan sheets x24 (48 sheets), kit card, Week 8 kit lists 5 min · reprographics

THE WEEK BEFORE

- [ ] Allocate options, print 10 option cards, make Evidence 6 and 7 bags 30 min · teacher
- [ ] Evidence 3 bags: unopened Week 4 bag, or about 3 tsp baking soda each 5 min · technician
- [ ] Note ink strips N: Week 3 spares or new with pen B, in clear sleeves 5 min · teacher
- [ ] Laminate the 10 option cards (optional) optional 10 min · technician
- [ ] Collect reference soils A to D and papers P1 to P4 (optional) optional 45 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Set out group kits, class box items; Evidence 6 and 7 on teacher desk 5 min · teacher

AFTER THE LESSON

- [ ] Clean-up: bags and N strips to teacher, keep option cards, wipe boards 3 min · students
- [ ] Read plan sheets: check risk tables, one comment each, read exit tickets 30 min · teacher
- [ ] Turn the six kit requests into Week 8 tray lists 10 min · teacher

PRINT

- Plan sheet x24 (4 A4 pages: 2 sheets each, double-sided)
- Option cards x10 (1 sheet each, single-sided: 6 group, 4 spare)
- Teacher copies: kit card (A5), Week 8 kit lists by option (4 A5)

GET OR MAKE (NEW THIS LESSON)

- Sealed evidence bags, up to 2 each: Evidence 6, 7, 3
- Note ink strips N x2 (Week 3 spares first)
- Small zip bags x6
- Sticky labels x6
- Clear plastic sleeves x2
- Lab paper towel sheets x2 (from the lab's own roll)
- Baking soda 6 tsp, if no sealed Week 4 bag (optional)
- Coffee-filter strips 2 cm x 10 cm x2 (optional)
- Mini whiteboards x24 (optional)
- Dry-wipe pens x24 (optional)
- Laminating pouches A4 x10 (optional)
- Reference soils A to D, papers P1 to P4 (optional)

Have ready from earlier weeks: Class box, Case files x24 (Week 4 results grid), Scientist's Logbooks x24, Role card sets x6 (with buddy card), Sentence frame card sets x6, My Science Dictionary x24, Pencils x24, Pen B, for new N strips (optional)

YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 8 OF 10 · 40 MINUTES

Lab day 检测日

Doing a planned test carefully, repeating it and recording every result straight away (surprises too) makes a result trustworthy evidence.

- I can carry out our plan safely, repeat the test and record results with units in a table.
- I can say what our result matches, how reliable it is and what it does or does not show.

repeat 重复 · reliable 可靠的 · anomaly 异常值 · record 记录 · results table 结果表格 · match 匹配 · settle 沉降 · absorb 吸收 · solvent 溶剂 · mixture 混合物

THE LESSON

Do Now	3	Safety Officer reads kit card aloud; group checks tray against kit request; roles confirmed.
Hook	2	Jury's three questions: repeated? references tested the same way? contaminated?
Learn	3	Safety slide, goggles on; record straight away, units, one line through, circle anomalies.
Do	24	Pairs run parallel repeats, scene sample last; record on sheet; photos 18 to 21; tidy from 21.
Think and Reflect	8	Worksheet; Reporter round, 20 seconds per group; evidence log row 8; exit ticket at the door.

Hands-on: Groups run their Week 7 plan (Soil, Paper, Ink 2 or Powder 2) as two pairs doing parallel repeats.

Collect or assess: Recording sheet and worksheet (strand 3, part of strand 4) into case files; evidence log row 8; exit ticket; photos.

SAFETY

- Goggles for vinegar, iodine, shaking soil bottles; lids tight; plastic bottles only
- Never taste; Powder 2 allergy check (wheat, corn): an allergic student is Recorder
- Soil: cover cuts or wear gloves, wash hands, never down the sink; iodine by dropper

WATCH OUT

- Check every tray before students arrive: with 24 Do minutes, a missing item costs a group its repeat.
- No rubbing out: one line through, circle anomalies. A match is weak or supporting evidence, never proof.

Language: Frame: 'The scene sample matches ___ because ___.' Traps: repeat vs replace, reliable vs accurate. Chinese welcome.

Next week: Week 9: soil bottles stay labelled on the windowsill (look, do not open); keep photos, board notes, strips.

PREP FOR ONE CLASS

before the lesson about 205 min (115 could be done by someone else); after about 78 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Print job 1: worksheets (48 sheets), run sheet A4 and A3, backup cards
  5 min · reprographics
- [ ] Buy or request consumables; ask canteen for napkins; check filters, tape
  15 min · teacher
- [ ] Collect, rinse 20 identical 300 ml plastic screw-lid bottles per Soil class
  15 min · technician

THE WEEK BEFORE

- [ ] Once options are allocated, print recording sheets x10 and kit cards x10
  5 min · reprographics
- [ ] Write a tray list per group from Week 7 plans; note each class's split
  10 min · teacher
- [ ] Gather prep-room kit, check pens A to D; bag soils A to D and Evidence 6
  50 min · technician
- [ ] Soil: if very wet, dry it on newspaper overnight (optional) optional
  5 min · technician
- [ ] Ink 2 pen pre-test (salt water, vinegar); keep dried strips as answer key
  30 min · teacher

THE DAY BEFORE

- [ ] Powder 2 pots (allergy check first); Ink 2 strips and vinegar; backup cards
  35 min · teacher
- [ ] Paper: make Evidence 7, cut and label P1 to P4, colour the water
  15 min · technician
- [ ] Paper: pre-cut strips if the class is slow with scissors (optional) optional
  10 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Fill Soil pots, set out bottles and one checked tray per group
  25 min · technician

AFTER THE LESSON

- [ ] Wash, restock trays; stick in Ink 2 strips; empty soil bottles after Week 9
  30 min · technician
- [ ] Save photos, board notes; read and file work; photograph soil bottles again 45 min · teacher

PRINT

- Worksheet x24 (3 pages on 2 sheets: pages 1 and 2 double-sided)
- Recording sheets x10 (1 sheet, double-sided; 12 if before allocation)
- Kit cards x10 (A5, or 2 per A4 sheet; 4 are for the teacher's file)
- Run sheet A3 enlargement x1 (wall)
- Teacher copies: run sheet A4, backup cards (1 sheet, cut into 4)

GET OR MAKE (NEW THIS LESSON)

- Identical 300 ml plastic screw-lid bottles x20
- Labels x39, markers x2, A4 paper x4, soil bags x5
- Filter strips 2 cm by 10 cm x50, plus spares
- Small plastic cups x44; stirring sticks x22
- Timers x5, scissors x4, bucket x1, extra trays x2
- Teaspoons x3, tablespoons x10, flat sticks x2
- Water jugs x4; measuring jugs or cylinders x4
- Small bottles for Ink 2 vinegar x2 (about 60 ml each)
- Gloves, cloths, paper towels; camera or tablet x1
- White vinegar 180 ml, plus a little for the pre-test
- Baking soda 12, salt 7, cornflour 2, flour 2 level tsp
- Tissues x2, canteen napkins x2, food colouring 6 drops
- Make: soils A to D; Evidence 3, 6 and 7 (x2 each)
- Make: P1 to P4 x2, Powder 2 sets x2, N strips x14
- Make: pre-test strips x6, coloured water 600 ml
- Optional: iodine dropper bottles x2, newspaper

Have ready from earlier weeks: Class box, case files, logbooks, Week 7 plan sheets and option cards, Role cards x6 sets; dictionaries, Goggles x20, magnifiers x5, rulers x7, Trays x6, tiles x4, droppers x2, tape x4, Pots x23, spoons x10, black card x2, Pens A to D x2 sets; kitchen roll, Pencils: 24 writing, 20 for strips

YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 9 OF 10 · 40 MINUTES

Build the case 构建证据链

No single clue proves who took the notebook: rank evidence by how many suspects it fits, set aside contaminated evidence with a checked reason.

- I can rank evidence as strong, supporting or weak, and explain why the puddle is set aside.
- I can write a CER conclusion with a confidence level and explain how tunnel vision misleads.

strong evidence 有力证据 **supporting evidence 支持性证据** **weak evidence 薄弱证据** **contamination 污染** **tunnel vision 隧道视野** **confirmation bias 确认偏误** **claim 主张** **reasoning 推理** **confidence 把握** **expert witness 专家证人**

THE LESSON

Do Now	3	Count each suspect's ticks in the suspect tracker: nobody matches everything. Slides 2 to 3.
Hook	2	Week 5 Ideas Parking Board photo: students find their note; today we check them. Slide 4.
Learn	5	Strong fits one suspect, supporting two, weak three or four; a combination can be strong.
Do	18	Build A3 board (7 min); cleaning log card, set Evidence 4 aside (4 min); own CER (7 min).
Think and Reflect	12	Cartoon and parking board sort (3); roles and frames (4); self-assessment (4); exit ticket (1).

Hands-on: No practical: groups build an A3 evidence board from evidence cards, then set aside the contaminated puddle.

Collect or assess: CER and photographed board: rubric strands 4 and 5, into the case file. Collect self-assessment slips and exit tickets.

SAFETY

- No practical hazards beyond normal classroom rules.
- Soil settling bottles stay sealed on the windowsill: look only, do not open.

WATCH OUT

- Keep cleaning log cards sealed in the envelope until Do minute 7, then give a full minute of quiet.
- Never confirm the culprit, even if every group says Dr Huang; nobody reads a model conclusion aloud.

Language: Chinese welcome in talk; English key words in writing. Traps: contamination/contaminated, confidence/confident.

Next week: Before Week 10: draft strands 1 to 5; return case files, frames. Print Week 10 set; optional prop to freezer.

PREP FOR ONE CLASS

before the lesson about 68 min (23 could be done by someone else); after about 108 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book the Week 9 print job with reprographics (all classes) 5 min · teacher

THE WEEK BEFORE

- [ ] Read Case Bible Section 9 and the answer key (all classes) 15 min · teacher
- [ ] Collect and count the printed worksheets, boards, frames and slips 5 min · reprographics
- [ ] Check stock; print and cut 6 card packs; cut 24 slips and 6 badges 18 min · technician
- [ ] Print 6 cleaning log cards, seal in envelope; decide roles, number badges 8 min · teacher
- [ ] Slide 4 photo; photograph soil bottles; gather Week 8 photos, notes 12 min · teacher

THE DAY BEFORE

- [ ] Optional: print the concept cartoon and put it up (optional) 3 min · technician
- [ ] Optional, if time is short: pre-fill 'Matches' lines on the 6 card packs (optional) 15 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out case files and resources; teacher desk in hand-out order 5 min · teacher

AFTER THE LESSON

- [ ] Photograph every evidence board 5 min · teacher
- [ ] Clean-up: work into case files; collect them, slips, tickets; badges to box 3 min · students
- [ ] Empty soil bottles on a flowerbed or into a bucket, never the sink 10 min · technician
- [ ] Read exit tickets and case files; draft strands 1 to 5 before Week 10 90 min · teacher

PRINT

- Worksheet x24 (2 sheets each, double-sided)
- Evidence card pack x6 (1 A4 card sheet each, cut into 9 cards)
- Evidence board x6 (A3 landscape)
- Cleaning log card x6 (A5)
- Statement frame x6 (A3); A4 alternative x24 (optional)
- Self-assessment slip x24 (12 A4 sheets, cut in half)
- Expert witness badges x6 (1 A4 sheet, cut out)
- Teacher copies: kit card, enlarged model evidence board
- Concept cartoon x1 (A4 landscape) (optional)
- Week 5 Ideas Parking Board photo print x1 (optional)

GET OR MAKE (NEW THIS LESSON)

- Blu Tack or glue sticks x6
- Envelope x1
- Camera or phone x1

Have ready from earlier weeks: Class box, Case files x24, Pencils and dictionaries x24, Role cards and frame card sets x6, Week 8 results photos and board notes, Soil bottles from Week 8 (windowsill), Week 5 Ideas Parking Board (optional)

YEAR 8 · SCIENCE DETECTIVES 科学侦探 · WEEK 10 OF 10 · 40 MINUTES

Court of Science 科学法庭

Scientists defend conclusions with evidence, answer hard questions honestly (including what evidence does not show) and accept the result of checking.

- I can explain one piece of evidence: what we tested, how, what it shows and how sure we are.
- I can answer or ask a question about evidence and reach a verdict on the strength of all of it.

court 法庭 · jury 陪审团 · verdict 裁决 · expert witness 专家证人 · testimony 证词 · evidence 证据 · confidence 把握 · case closed 结案

THE LESSON

Do Now: skills check	15	Same paper as Week 1, worked alone; 5-minute warning, collected at once; never shortened.
Hook	1	Teacher becomes judge; experts put on badges; juries take question cards and jury notes.
Do: Court of Science	14	Six 90-second talks in evidence order, one jury question each; jury notes, feedback slip.
Think and Reflect: verdict	5	Exit ticket verdict (30 seconds), vote; teacher reads reveal card, 概要 lines; case closed.
Think and Reflect: survey	5	Interest survey (3 minutes); thanks and certificates, or thanks if they come later.

Hands-on: No practical: groups give 90-second expert witness talks, answer a jury question; the class votes a verdict.

Collect or assess: Skills check (out of 20, vs Week 1); interest survey; strand 6 talk record; case files (strands 1 to 5); exit ticket.

SAFETY

- No practical hazards beyond normal classroom rules.
- Optional frozen notebook: cold and wet; pass it round in its bag, not out of it.

WATCH OUT

- Skills check as in Week 1: unaltered paper, same access, dictionaries away, never shortened.
- Talk past 2 minutes: thank, move on. Short of time: 75-second talks, 3 questions; never skip verdict.

Language: Frames on the stand; 'My verdict is ___ because ___.' Band 1 sits by language buddy; Chinese answers count for Core.

Next week: Last lesson. Then: mark, enter tracker, finalise strands, apply award rules; certificates within a week.

PREP FOR ONE CLASS

before the lesson about 62 min (28 could be done by someone else); after about 333 min (some times estimated)

2 OR MORE WEEKS BEFORE

- [ ] Book Weeks 9 and 10 printing (once); agree certificate ceremony with lead 10 min · teacher

THE WEEK BEFORE

- [ ] Book Week 1 reader or support for the skills check; make '15 seconds' card 7 min · teacher
- [ ] Send print job (only designs rebuilt after the language pass) 5 min · reprographics
- [ ] Cut question cards (6 sets of 18), surveys, slips; staple checklists 23 min · technician

THE DAY BEFORE

- [ ] Label slips by group; names in talk record; pick 概要 reader; check buddies 10 min · teacher
- [ ] Optional: freeze an old notebook in a freezer bag overnight optional 3 min · teacher
- [ ] Optional: print certificates for awards already fixed, if given today optional 20 min · teacher

ON THE DAY, BEFORE THE LESSON

- [ ] Set out: papers face down, table packs, witness stand, timer, clipboard 7 min · teacher

AFTER THE LESSON

- [ ] Clean-up: skills checks first, surveys, case files; badges, cards in box 3 min · students
- [ ] Sort feedback slips and reveal cards into group case files 10 min · technician
- [ ] Mark skills checks out of 20; enter tracker; finalise strands; check awards 145 min · teacher
- [ ] Optional: enter question marks on the tracker's Question_analysis tab optional 15 min · technician
- [ ] Print and sign 25 certificates; skills reports; photos; programme summary 175 min · teacher
- [ ] Optional: return Week 1 and 10 skills checks after marking and entry optional 5 min · teacher

PRINT

- Skills check x24 (3 sheets each, double-sided, as Week 1)
- Interest survey x24 (half A4, 2 per sheet, cut: 12 sheets)
- Jury notes sheet x24 (1 sheet each, double-sided)
- Peer feedback slips x24 (A6, 4 per sheet, cut: 6 sheets)
- Final case file checklist x24 (A5, 2 per sheet, cut: 12 sheets)
- Reveal card x7 (1 sheet each: 1 teacher, 1 per group)
- Jury question cards x6 sets (2 A4 card sheets each, cut)
- Teacher copies: run sheet, strand 6 talk record (2 sheets), kit card
- After marking: certificates x25 (A4 card), skills reports x24
- Extra expert witness badge sheets x3 (6 per sheet, cut) (optional)

GET OR MAKE (NEW THIS LESSON)

- Timer visible to the class x1 (on-screen works)
- Clipboard x1
- Stapler and staples x1
- '15 seconds' card x1 (hand-written)
- Old notebook x1 and freezer bag x1 (optional)
- Staff freezer space overnight (optional)

Have ready from earlier weeks: Case files x24 (with Week 9 frames), Expert witness badges (6 from Week 9), Class box, Pencils x24 (Week 1), Role cards, frame cards (6 sets each), Language cards x24, wall posters x4, Dictionaries x24, Logbooks x24, Teacher copies: mark scheme, rubric